

High Cr can cause hardening within the bulk matrix but too low of a concentration reduces oxide resistance. Other composition optimizations are possible through adjusting Si, Ni, etc. Corrosion resistance, strength, and ductility must be maintained. Additionally, stabilizing the F/M structure so it doesn't become δ -ferrite is desirable. Preventing swelling is important (maintaining the structure & including oxides as fine particles aids in this.) Finally from a simple utility standpoint, avoiding materials with α is important. 10/10 89

Austenitic steels are primarily FCC and are far more "soft" of a matrix than the rigid bcc structure of ferrite. Additionally, the many slip planes of FCC and the high APF assure that irradiation enhanced creep occurs and there is not much intragranular space in the matrix to accommodate gas leading to the formation of defect clusters rather than uniform distribution. BCC has γ_0 a lower APF therefore decreasing the driving force of defect diffusion. Less voids & more strength makes F-steels swell less. - missed some key points

DBTT is a phenomenon which occurs in BCC materials due to the presence of thermally activated slip planes. As T decreases the material transitions to a more brittle type. As dose is accumulated and irradiation damage impedes dislocation motion through slip planes, less at especially high T, an upward shift in the DBTT is observed (this has been specifically noted in 9Cr-F/M steel). The formation of $M_{23}C_6$ and monocarbide precipitates which accumulate at grain boundaries particularly increase the irradiation embrittlement through arresting grain boundary sliding. 10/10

Fukushima revealed the concerns of the exothermic oxidation reaction which occurs with Zircaloy when exposed to high temperature team. Thus, accident tolerant fuels are of particular interest. FeCrAl boasts high strength, good corrosion resistance, and maintains favorable properties even at high burnup levels. The ratio of Cr/Al is of particular interest in this alloy. Too high of a Cr/Al ratio leads to excessive hardening within the matrix while too low can lead to Al intermetallics. Both extremes result in undesired precipitate phases.

- Specifically Steam oxidation resistance
 - Too low insufficient corrosion resistance
- 6/8

articles a oxide dispersion strengthened steels which situate at grain boundaries and prevent sliding and increase the strength of the material.

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- detect sinks

dispersed uniformly

1) The main advantage of Ni alloys (>28% Ni alloys) is the corrosion resistance. They also have less swelling than SS and have high strength. The drawbacks include the $\text{Ni-Si}(\text{n}, \alpha)$ reaction which produces high volumes of He gas, which can cause embrittlement and swelling. This is a concern for high burnup behavior.

- high T mech. properties

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2) Both are dispersion types, UAl_x fuel has the feature of UAl_x production with Al which increases "storage" volume for fission gases while U₃Si does not. U-Si has a higher fissile density which allows for more potential neutron/power production. Another drawback is U₃Si creates an interaction layer with the Al cladding which can degrade, cause pore connection and blistering, and ultimately cause cladding failure. U₃Si swells at a higher rate than U₃Si₂ as it is a softer material. Because of this, the stoichiometry is different for plate type fuel and rod type U-Si fuel. More swelling can be accommodated in rod-type.

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3) The γ phase of U-Mo is dominant in-reactor due to the irradiation effects which alter the most energetically favorable configuration of the atoms. As the defects are accumulated, this structure is able to accommodate the dispersed interstitials and vacancies produced from PKA Frenkel defects.

- kind of

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4) While using the hot isostatic press method to produce the fuel foils for monolithic U-Mo fuels, not raising the temperature as high as to melt the fuel or allow high temperature induced diffusion and redistribution of the fissile density in the plate is important. The way U-Mo fuel is sheets of varying compositions makes keeping the liquidus line for the composition where that T is lowest in mind so the fuel fabrication is as desired. Moly-lean areas can cause unfavorable macroscopic properties so lower processing temperatures can assure this does not occur.

- you sure wrote a lot. but didn't really answer the question...

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1) The fissile density directly affects grain refinement which directly affects FG bubbles. At low fissile density, little effect to the material is observed (N_{2e27} f/cm). Medium, we start to see grain refinement. High, nearly the whole matrix's grains are refined. This increases the grain boundary surface energy drawing more bubble production. At especially high burnups, this can lead to delamination (N_{9e27}).

$\%10$ - fission gas supersaturation forms + is destroyed

2) The IL in U-Mo fuels can become quite thick and induces pores and can degrade the cladding and fuel performance. Including a Zr buffer layer is an option to reduce the diffusion. Adding Si or other alloying materials to the U-Mo has been considered. This way the thickness of the γ interaction layer may be reduced and delamination from the cladding won't occur.

3) Al features good ductility, less than 1% of swelling under irradiation, good strength after mechanical strengthening treatment, and low neutron interaction. As a non-removable, structural component in a research reactor, the primary concern is radiation hardening and embrittlement, but its good point defect mobility promotes large amounts of recombination at high burnup. However, Al has a very low melting point, making it not particularly applicable as a structural element in power generation applications such as LWRs. Also, at extremely high fluence, its recombination, self-healing will be far less effective leading to unfavorable hardening.

Fun fact: All the structural elements of the PULSTAR are made of Al except for the SS screws which hold them together. Note: Al in LWR's will probably be higher load bearing as well making it not a great candidate as it is not as strong as other metals.

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